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THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

SIGNALS AND SYSTEMS

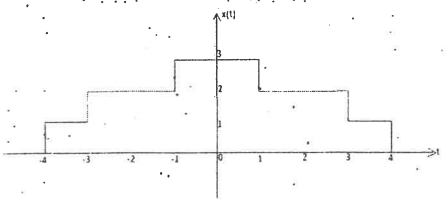
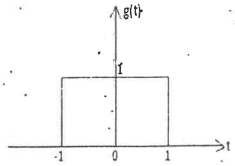
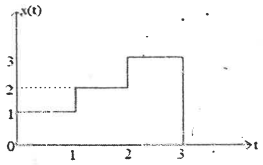
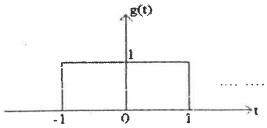
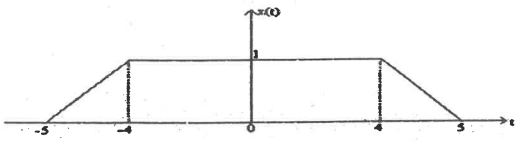
Time: 3 Hours **Maximum Marks: 100**

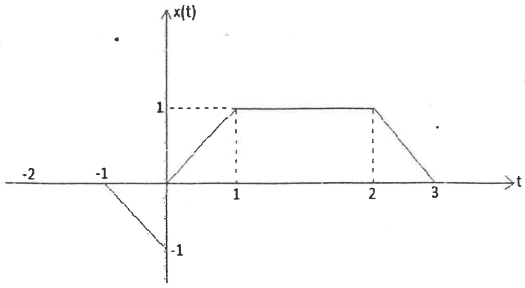
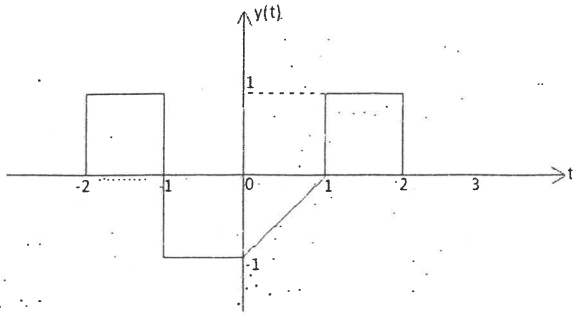
Instructions: i. Missing data may be suitably assumed.

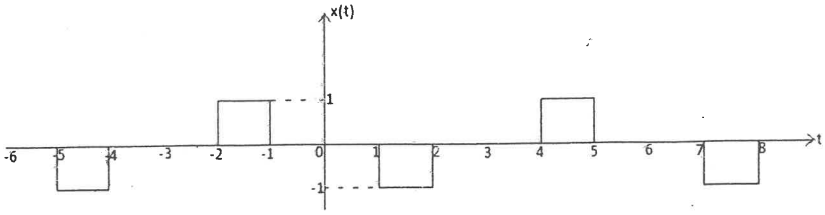
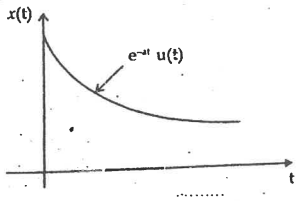
ii. Answer All Questions

PART-A **5 X 2=10**

1.	**a.**	Determine the fundamental period of the signal $x(t) = sin(5\pi t) + cos(3\pi t)$	L3	CO1	**02**
	b.	Find the step response for the LTI system represented by the impulse response $h(n) = \left(\frac{1}{2}\right)^n u(n)$	L3	CO2	**02**
	c.	State convolution and modulation properties of continuous time Fourier Series.	L3	CO3	**02**
	d.	State sampling theorem and Nyquist rate criteria.	L2	CO4	**02**
	e.	Find the Z Transform of $x[n] = u[n - no]$.	L2	CO4	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Find the even and odd components of each of the following signals: i. $x(t) = 4 + 3t + 5t^2 + 8t^3 + 9t^4$ ii. $x(t) = cos\, wt + 2sin\, wt + 3cos^2\, wt$	L3	CO1	**05**
		OR			
	b.	Check whether the following signal x(t) is an energy or power signal and find the corresponding value. $x(t) = \begin{cases} t; & 0 \le t \le 2 \\ 0; & Otherwise \end{cases}$	L3	CO1	**05**
3.	**a.**	Evaluate the continuous time convolution integral of $x(t) = u(t)$ and $h(t) = u(t-2)$	L3	CO2	**05**
		OR			
	b.	Determine the convolution sum of two sequences $x(n) = \{1,2\}$ and $h(n) = \{3,4\}$	L3	CO2	**05**
4.	**a.**	State any five properties of DTFS.	L2	CO3	**05**
		OR			
	b.	Find the Fourier coefficients of $x(t) = sin\, wt$ using Fourier Series.	L3	CO3	**05**
5.	**a.**	Find DTFT of the signal $x[n] = (0.5)^{n+2} u[n+2]$.	L2	CO4	**05**

		OR			
	b.	Find the Fourier transform of a rectangular pulse described below $x(t) = \begin{cases} A\,; & -T/2 \le t \le T/2 \\ 0\,; & otherwise \end{cases}$	L3	CO4	**05**
6.	**a.**	Determine Z transform, ROC pole and zero location of $x(n) = \left(\frac{1}{2}\right)^n u(n) + \left(-\frac{1}{3}\right)^n u(n)$	L3	CO4	**05**
		OR			
	b.	Prove the following properties of z-transform mentioning ROC **i.** Time shifting property **ii.** Differentiation in Z Domain	L3	CO4	**2+3**
7.	**a.**	A continuous time signal $x(t)$ and $g(t)$ are shown in figure7a(1) and figure 7a(2) respectively. Express x(t) in terms of g(t).  Figure 7a(1)  Figure 7a(2)	L3	CO1	**05**
		OR			
	b.	Two signals $x(t)$ and $g(t)$ are shown in figure 7b(1) and 7b(2) respectively. Express the signal $x(t)$ in terms of $g(t)$.  Figure 7b(1)  Figure 7b(2)	L3	CO1	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	The trapezoidal pulse x(t) shown in below figure 8a If it is applied to the differentiator defined by $y(t) = \frac{d}{dt}x(t)$  Figure 8b **i.** Find the resulting output y(t) of the differentiator **ii.** Find the total energy of y(t). **iii.** Find the total energy of x(t).	L3	CO1	**10**

		OR			
	b.	For the continuous time signals x(t) and y(t) shown in the below figure 8b(1) and 8b(2), Sketch $(i) z(t) = x(2t)\, y(2t + 1)$ $\quad$ $(ii)\, z(t) = x(t)\, y(-t-1)$  Figure 8b(1)  Figure 8b(2)	L3	CO1	**10**
9.	**a.**	Evaluate the continuous-time convolution integral given below and compute the output of the system y(t). $y(t) = e^{-2t}\,\mathrm{u(t)} * \mathrm{u(t+2)}$	L3	CO2	**10**
		OR			
	b.	Check whether the following systems are i) Memoryless, ii) Causal and iii) Stable, for each of the impulse response listed below: **i.** $h(t) = e^{2t}.\mathrm{u(t-2)}$ **ii.** $h[n] = \left(\frac{1}{3}\right)^n \mathrm{u(n+1)}$	L3	CO2	**10**
10.	**a.**	Obtain the frequency response and the impulse response of the system described by the difference equation: $y(n) + \frac{1}{4}y(n-1) - \frac{1}{8}y(n-2) = x(n) - \frac{3}{4}x(n-1)$	L3	CO2	**10**
		OR			
	b.	Find the transfer function and impulse response of the system described by $y(n) - 1/2\, y(n-1) = 2\, x(n-1)$	L3	CO2	**10**

11.	**a.**	Find the Fourier Series co-efficient of the periodic signal x(t) shown in the figure 11(a)  figure 11(a)	L3	CO3	**10**
		OR			
	b.	Find the Fourier transform and sketch magnitude and phase spectrum of the following signal shown in figure 11(b).  Figure 11(b)	L3	CO4	**10**
12.	**a.**	Find the DTFT of a signal **i.** $x(n) = \left(\frac{1}{2}\right)^{n+2} u(n)$ **ii.** $x(n) = \left(\frac{1}{4}\right)^{n} u(n-2)$	L3	CO4	**10**
		OR			
	b.	State and prove the following properties of DTFT **i.** Frequency Differentiation **ii.** Convolution **iii.** Modulation	L2	CO4	**10**
13.	**a.**	Solve the difference equation using Z-transform method $$y(n) - \frac{3}{2}y(n-1) + \frac{1}{2}y(n-2) = \left(\frac{1}{4}\right)^{n} u(n)$$ with initial conditions $y(-1) = 4$ *and* $y(-2) = 10$.	L3	CO4	**10**
		OR			
	b.	Suppose x(z) is given by $$x(z) = \frac{z(z^2 - 4z + 5)}{(z-3)(z-2)(z-1)}$$ find x(n) for the following ROCs **i.** $2 < \lvert z \rvert < 3$ **ii.** $\lvert z \rvert > 3$ **iii.** $\lvert z \rvert < 1$	L3	CO4	**10**